1 A capacitor is connected to a reed switch, a d.c. power supply of negligible internal resistance and a resistor as shown in Figure 1.1.

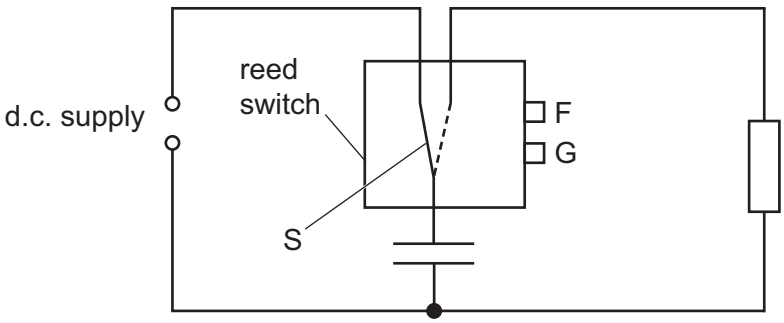

Figure 1.1

The reed switch is a device that contains a two-way switch S that connects the capacitor either to the d.c. supply or to the resistor.

An alternating current is supplied to terminals F and G of the reed switch. This causes switch S to vibrate. The frequency *f* of switch S is the same as the frequency of the alternating current.

For half of one cycle of the alternating current, the capacitor is charged from the d.c. supply. For the other half of the cycle, the capacitor discharges through the resistor. The mean current in the resistor is I.

The capacitor is created from two circular metal plates in air. The plates each have area *A*. The plates are separated by a small distance *h*.

It is suggested that I is related to *h* by the relationship

$$\frac{KA}{h} = \frac{I}{fE}$$

where *E* is the electromotive force (e.m.f.) of the d.c. supply and *K* is a constant.

Plan a laboratory experiment to test the relationship between I and *h*.

Draw a diagram showing the arrangement of your equipment.

Explain how the results could be used to determine a value for *K*.

In your plan, you should include:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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